SHAH SANJIDA RAHMAN NITU



Computer Science & Engineering | Machine Learning & Astronomical Data Research

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PROFILE

Final-year Computer Science and Engineering undergraduate at RUET working on machine learning for astronomical time-series data. Current research applies unsupervised learning, self-supervised representation learning, anomaly detection and uncertainty estimation to TESS and Kepler exoplanet data, with emphasis on reliable predictions and physically faithful preprocessing. Brings proven leadership as Project Manager of SpaceShare Solutions, a pre-seed-funded startup (UIHP, RUET), and as an executive of two university organisations, alongside experience mentoring student researchers and hosting 10+ professional events.

EDUCATION

B.Sc. in Computer Science and Engineering — Rajshahi University of Engineering & Technology (RUET) Expected Mar 2027

CGPA: 3.34 / 4.00 · Focus areas: machine learning, deep learning, data analysis

RESEARCH EXPERIENCE

Uncertainty-Aware Machine Learning for Reliable Exoplanet Candidate Validation IN PROGRESS

- Developing a validation approach that pairs each classification with a calibrated confidence estimate, so that low-certainty candidates can be flagged for expert review instead of being accepted automatically.
- Studying model confidence and uncertainty-estimation methods on labelled Kepler exoplanet data.

A Gap-Aware and Transit-Preserving Preprocessing Framework for Unsupervised Analysis of TESS Light Curves IN PROGRESS

- Designing a preprocessing pipeline for TESS photometry that handles observation gaps and sector boundaries without distorting or removing transit signals.
- Intended as the data foundation for downstream clustering and anomaly-detection models.

Unsupervised Clustering Benchmark for TESS Light Curves IN PROGRESS

- Benchmarking unsupervised clustering of TESS light curves to test whether discovered clusters are enriched for known planet-hosting stars.
- Built a development set of 5,000 randomly sampled stars per sector across TESS Sectors 1–5; progress reported regularly to faculty supervisor.

Unsupervised Discovery of Potentially Uncatalogued Exoplanets in TESS Light Curves Using Self-Supervised Representation Learning and Anomaly Detection IN PROGRESS

- Exploring self-supervised representation learning to embed TESS light curves and anomaly detection to surface unusual, potentially uncatalogued transit-like signals.

All projects are ongoing under faculty supervision; no results or publications are claimed at this stage.

DATASETS

- **TESS light curves** — NASA MAST archive, bulk downloads (Sectors 1–5 development set)
- **Kepler exoplanet dataset** — tabular CSV in two sizes: standard and extended

TECHNICAL SKILLS

Programming	Python
Machine Learning	Supervised classification, unsupervised clustering, anomaly detection, uncertainty estimation
Deep Learning	Neural networks, advanced deep learning, self-supervised representation learning; TensorFlow
Data	Astronomical time-series (light curve) preprocessing, exploratory data analysis
Tools	Google Colab, Git & GitHub

LEADERSHIP EXPERIENCE

Vice President, Monitoring & Evaluation — RUET Computing Society Sep 2026 – Mar 2027

- Oversee monitoring and evaluation of the society's programmes and activities.

Vice Chair, Program Management — RUET CORE (Collaborative Opportunities in Research and Education) Sep 2026 – Mar 2027

- Coordinate the planning and delivery of research and education programmes for students.

Research Mentor — RUET CORE

- Guide fellow students in starting and carrying out research projects.

Project Manager & Team Leader — SpaceShare Solutions (student startup)

- Led the team in refining the startup concept and developing a scalable business model.
- Secured pre-seed funding through the University Innovation Hub Program (UIHP) at RUET under the DEIED Project (announcement).

HONOURS & COMPETITIONS

Pre-Seed Funding Recipient — University Innovation Hub Program (UIHP), RUET

Accomplished Competitor — University Physics Competition 2025

Competitor — University Physics Competition 2026

- Team of three; modelling Problem B (penalty-kick trajectory prediction).

TRAINING

AI/ML Training Program — Phitron

- Completed Python for Machine Learning, Machine Learning and Deep Learning; selected for a three-month thesis track on NLP and Large Language Models (announcement).

VOLUNTEERING & PUBLIC SPEAKING

Volunteer — WordCamp Rajshahi 2026 2026

- Supported a WordPress and open-source community event across Contributor Day and Conference Day.

Anchor — National Android Development Bootcamp

- Hosted a national technical bootcamp sponsored by bdapps in collaboration with MIAKI (event post).

Event Host & Public Speaker

- Hosted 10+ professional and organisational events, including technical programmes, university club activities, orientations and corporate events.